Houjiang Liu

Ph.D. Candidate

Contact Information

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Education

University of Texas at Austin, School of Computing, 2021– Expected May 2027
Ph.D. Candidate of Information Science
Dissertation: *From Automation to Agency: Evaluating Computational Support for Human Research Ideation*
Committee: Matthew Lease (advisor), Jacek Gwizdka, Ahmer Arif (Columbia University), and Aniket Kittur (Carnegie Mellon University)

Northeastern University, College of Arts, Media and Design, June 2020
M.F.A. in Experience Design
Dissertation: *Enriching Gifts, Experience Design to Understand Human Relationships* [P5]
Committee: Miso Kim (advisor), Kristian Kloeckl, & Meryl Alper

Jilin University, College of Physics, July 2015
B.S. in Applied Physics

Funding & Fellowship

Research Grant Fund, Cosmos Institute, 2026 — $5,000, AI and Human Autonomy
E. Glynn Harmon Excellence Fund, University of Texas at Austin, 2026 — $1,000
Outstanding Research Fellowship, University of Texas at Austin, 2025-26 — $38,000
Summer Research Fellowship, University of Texas at Austin, 2023 — $10,832
Summer Research Fellowship, Northeastern University, 2019 — $5,000
Graduate Scholarship, Northeastern University, 2018–20 —50% tuition remission
Undergraduate Merit Scholarship, Jilin University, 2012–13, 2015 — with direct admission to graduate program

Honors & Awards

Best Paper Honorable Mention, ACM CSCW, 2024 — top 4% of submitted papers
Professional Development Award, University of Texas at Austin, 2023, 2024, 2025
Information is Beautiful Awards — Longlist, Data Visualization Society, 2022
Dean's Project Award for Research, Scholarship, and Creative Activity, Northeastern University, 2020 — $2,000
First Design Award, Health x Design Challenge, Mad*Pow, 2019 — $5,000
Innovative Design Award, Reinventing Tencent Map, Tencent, 2014
Second Prize, Math Modeling, Jilin University, 2012

Research

*Authors with equal contributions are marked with **

Manuscripts Under Peer-Reviewed [U]

[U2] Jiaxin An, **Houjiang Liu**, and Yan Zhang. "Conversational Agents for Older Adults' Health Information Seeking: A Systematic Literature Review." Under review, ACM CHIIR 2027

[U1] **Houjiang Liu*,** Yujin Choi*, Sanjana Gautam, Gabriel Jaffe, Soo Young Rieh, and Matthew Lease.

"Who Owns Creativity and Who Does the Work? Trade-offs in LLM-Supported Research Ideation. Under revise and resubmit, *International Journal of Human-computer Interaction* [arXiv]

Peer-Reviewed Conference Proceedings and Journal [P]

[P13] **Houjiang Liu*,** Li Shi*, Yian Wong, Utkarsh Mujumdar, Dan Zhang, Jacek Gwizdka, and Matthew Lease.
"Reducing Confirmation Bias on Controversial Issues through LLM-Generated Multi-Persona Debates." *Information Processing & Management 64 (2): 105240*, 2027 [DOI, arXiv, Slides]

[P12] Sanjana Gautam, **Houjiang Liu,** Yujin Choi, and Matthew Lease.
"How Researchers Navigate Accountability, Transparency, and Trust When Using AI Tools in Early-Stage Research: A Think-Aloud Study." *Proceedings of the 9th ACM Conference on Fairness, Accountability, and Transparency, pp. 1494–1512, 2026,* Montréal, Canada, 2026. [DOI, arXiv]

[P11] **Houjiang Liu,** Jacek Gwizdka, and Matthew Lease.
"Exploring Multidimensional Checkworthiness: Designing AI-assisted Claim Prioritization for Human Fact-checkers." *Proceedings of ACM Human-Computer Interaction, 9 (7): CSCW292,* 49 pages, 2025. [DOI, arXiv, Code, Slides, Video]

[P10] **Houjiang Liu*,** Anubrata Das*, Alexander Boltz*, Didi Zhou, Daisy Pinaroc, Matthew Lease, and Min Kyung Lee.
"Human-centered NLP Fact-checking: Co-Designing with Fact-checkers using Matchmaking for AI." *Proceedings of ACM Human-Computer Interaction. 8, CSCW2, Article 423,* 44 pages, 2024. [DOI, arXiv, Slides]
Best Paper Honorable Mention (top 4%)

[P9] Anubrata Das, **Houjiang Liu**, Venelin Kovatchev, and Matthew Lease.
"The state of human-centered NLP technology for fact-checking." *Information Processing & Management, 60(2) :103219,* 2023. [DOI, arXiv]

[P8] **Houjiang Liu*,** Estefania Ciliotta Chehade*, William Albert, and Miso Kim.
"An Empirical Study of How Service Designers Use Metrics." *She Ji: The Journal of Design, Economics, and Innovation, 9(3) :402–427,* 2023. [DOI, Blog]

[P7] Mingzhu Li, Miso Kim, Liqing Huang, and **Houjiang Liu**.
"Designing for people you can't meet: Persuasive design to nurture seniors' participation in community services during the COVID-19 pandemic." *The Design Journal, 26(6) :934–955*, 2023. [DOI]

[P6] Sara Colombo, Estefania Ciliotta Chehade, Lucia Marengo, **Houjiang Liu**, Piero Molino, and Paolo Ciuccarelli.
"Design for Emergency: How Digital Technologies Enabled an Open Design Platform to Respond to COVID-19." *Interacting with Computers, 35(5) :650–665,* 2023. [DOI, Website, Blog, Video]

[P5] **Houjiang Liu** and Miso Kim.
"Is gift a token of gratitude or an imposition of identity? Facilitating positive

consequences of gift-giving with receiver-centred design." *Proceedings of the 10th International Association of Societies of Design Research (IASDR),* Milan, Italy, 2023. [DOI, Slides]

[P4] Jiabao Li, **Houjiang Liu**, Jilie Zeng, Di Wu, Ying Ding, and Alec McGail. "Shifting Winds: Gendered Structures of Academic Mentorship." *Proceedings of 2022 IEEE VIS Arts Program (VISAP), pp. 14–29,* ieeexplore, Oklahoma City, USA, 2022. [DOI, Website, Code]

[P3] **Houjiang Liu**, Ray Lc, Miso Kim, Concettina Cormio, and Meixin Yu. "Designing for Distance Nursing: Reconnecting Nursing Students with Senior Home Residents During COVID-19." *Proceedings of the 9th International Association of Societies of Design Research (IASDR)*, Hong Kong, 2022. [DOI, Slides]

[P2] **Houjiang Liu**, Miso Kim, Mingzhu Li, and Shruthi Lakshmi Narayan. "Technology, Autonomy, and Participation: Designing Community Games and Services to Enhance Older Adults' Technology Literacy." *Proceedings of the 9th International Association of Societies of Design Research (IASDR),* Hong Kong, 2022. [DOI, Slides]

[P1] **Houjiang Liu**, Miso Kim, Canqun He, and Tia Thomson. "Autonomy as a Design Principle: Service Design for the Technology Literacy of Older Adults." *Proceedings of the Cumulus Conference on Design Culture(s)*, Rome, Italy, 2021. [DOI]

Lightly Reviewed Extended Abstracts, Short Papers, & Posters [E]

[E4] Huimin Xu, **Houjiang Liu**, Yan Leng, and Ying Ding. "Adapting to LLMs: How Insiders and Outsiders Reshape Scientific Knowledge Production." *International Conference on the Science of Science and Innovation*, Boulder, USA, 2026. [arXiv]

[E3] **Houjiang Liu*,** Yujin Choi*, Sanjana Gautam, Gabriel Jaffe, Soo Young Rieh, and Matthew Lease. "Effects of Varying Control vs. Automation in Human-in-the-loop Research Ideation." *Amazon Research Day, Workshop on AI co-scientist: Accelerating Scientific Discovery through Intelligent AI Collaboration,* Palo Alto, USA, 2026. [pdf]

[E2] Anubrata Das, **Houjiang Liu**, Venelin Kovatchev, and Matthew Lease. "The Need for Human-centered Design in Fact-checking Research." *Proceedings of the 1st Information Processing & Management (IP&M) Conference*, Xiamen, China, 2022. [pdf]

[E1] **Houjiang Liu**. "Enriching Family Gifts: Reciprocity in Family Relationships." (Extended Abstract of C5). *NA PhD by Design Symposium*, New York University, USA, 2020. (Symposium canceled due to COVID19 shutdown)

Manuscripts [M]

[M1] **Houjiang Liu**, Yiheng Su, and Matthew Lease.
"Rhetorical XAI: Explaining AI's Benefits as well as its Use via Rhetorical Design" [arXiv]

Invited Talks

"Assessing Creativity vs. Efficiency Tradeoffs across Control Levels of LLM-Supported Research Ideation." *CosmicAI Biweekly Seminar*, February 2026.

"Exploring Multidimensional Checkworthiness: Designing AI-assisted Claim Prioritization for Human Fact-checkers." *iSchools Global Doctoral Seminar Series*, August 2025.

"Generating Multi-Persona Debates with LLMs to Explore Diverse Perspectives on Controversial Topics." *Kungfu.AI Guest Lecture*, May 2025.

"Generating Multi-Persona Debates with LLMs to Explore Diverse Perspectives on Controversial Topics."

- *Good Systems Annual Symposium, Shaping the Future of Ethical AI.* University of Texas at Austin, Spring 2025.
- *iSchool AI Showcase: Community-Engaged Research on AI and Society.* University of Texas at Austin, Fall 2024.

"An Exploration of Service Design Metrics." *Research, Innovation, Scholarship, Entrepreneurship (RISE), Virtual EXPO*. Northeastern University, 2021.

"Participatory Design for Senior Healthcare: Service Design to Improve Seniors' Health Literacy and Autonomy." *Research, Innovation, Scholarship, Entrepreneurship (RISE)*. Northeastern University, 2020.

Selected Design Exhibitions

"Biophilia: Mapping the Emotional Pulse of Urban Nature." *Information+ Conference*, Boston, 11/14-16, 2025 [Exhibition]

"Shifting Winds: Gendered Structures of Academic Mentorship." *IEEE VIS Arts Program (VISAP),* Oklahoma City, 10/16-21, 2022 [Exhibition]

"Data Sonification Archive." *2022 Design Research Week*, Boston, 4/25-29 [Website]

"How Food is Served in Early Care Center." *Northeastern University Visualization Consortium (NUVis) Demofest*. Boston, 2019.

Teaching Experience

Teaching Fellow, College of Arts, Media and Design, Northeastern University
ARTG2400 Interaction Design Principles, Spring 2021 – Instructor of Record
ARTG2401 Interaction Design Tools, Fall 2019, Spring 2020 – Instructor of Record

Paired undergraduate studio courses covering interaction design research, prototyping, usability testing, iterative design, and implementation of interactive web experiences. I developed and delivered course content, designed assignments and studio activities, and guided students through iterative project development and critique.

Teaching Assistant, College of Arts, Media and Design, Northeastern University
ARTG6330 Info Design Mapping Strategies, Fall 2020
Graduate course exploring mapping as a method for understanding and communicating relationships between information, place, and context. As a teaching assistant, I supported studio instruction, provide technical support, such as online data extraction and visualization, facilitated critiques, and provided feedback on students' design projects.

ARTG6900 Design for Dignity, Fall 2019
Graduate studio course examining the ethical dimensions of human-centered and service design through the concept of dignity. As a teaching assistant, I supported class activities and discussions and guided students through the development and critique of design projects.

Academic Service

Selected Reviewing (5 x Recognized for Outstanding Reviews)

- ACM CHI, 2026-27 *(1 outstanding review)*
- ACM CSCW, 2022–27 *(3 outstanding review)*
- ACM UIST, 2024, 2026 *(1 outstanding review)*
- ACM Collective Intelligence (CI), 2025
- ACM DIS, 2025
- *International Journal of Human-Computer Interaction* 2026
- *International Journal of Intercultural Relations* 2026
- *Information Processing & Management* 2026
- Design Research Society Conference (DRS) 2024

Student Volunteer

- ACM CSCW, 2024-25
- ACM International World Wide Web Conference (WebConf), 2023
- ACM SIGIR Conference on Human Information Interaction Retrieval (CHIIR), 2022
- User Experience Professionals Association Conference (UXPA) International, 2015

Ph.D. Representative, School of Information, University of Texas at Austin

- Research Committee 2024–26
- Master Studies Committee 2021–22

Professional Experience

Graduate Research Assistant, AI & Human-Centered Computing Group, University of Texas at Austin. 2021–present
Conduct interdisciplinary research on human–AI interaction, with a focus on creativity support, expert work, and responsible AI. Lead research activities across funded projects, including empirical study design, interactive system prototyping, data analysis, manuscript development, and project coordination.
Selected funded sources: UT Good Systems; CosmicAI NSF/Simons-supported research institute.

Design Researcher, Center for Design, Northeastern University. 2020–21

Conducted design research in service design and information visualization, applying qualitative and participatory design methods to investigate technology use and needs among older adults.
Selected funded sources: Center for Design Bridges Fellowship

Lecturer, College of Arts, Media and Design, Northeastern University. 2020–21
Taught interaction design and user experience courses, with an emphasis on design research, prototyping, and human-centered design methods.

User Experience Designer, SeetaTech, Beijing. 2017–18
Designed user experiences for AI-powered products, translating user research and design insights into product strategies, interaction flows, and interface designs in collaboration with product and engineering teams.

Interaction Designer, JD.com, Beijing. 2015-2017
Designed user experiences for large-scale e-commerce products across web and mobile platforms, conducting user research and developing interaction frameworks in collaboration with product and engineering teams.

Interaction Design Intern (2015 Spring, JD.com; 2013 Summer, Lenovo; 2012 Summer, Midea Group)